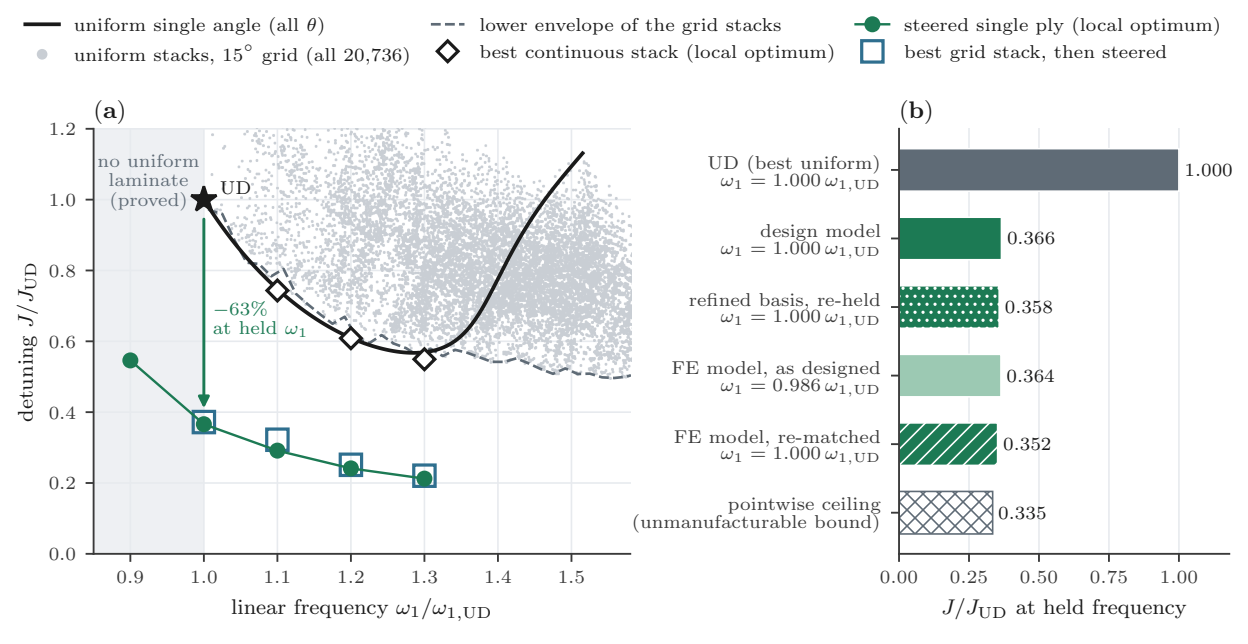

uniform single angle (all θ)
uniform stacks, 15° grid (all 20,736)
lower envelope of the grid stacks
best continuous stack (local optimum)
steered single ply (local optimum)
best grid stack, then steered
(a)
no uniform laminate (proved)
UD
−63% at held ω1
detuning J/J_UD
linear frequency ω1/ω1,UD
(b)
UD (best uniform) ω1 = 1.000 ω1,UD
1.000
design model ω1 = 1.000 ω1,UD
0.366
refined basis, re-held ω1 = 1.000 ω1,UD
0.358
FE model, as designed ω1 = 0.986 ω1,UD
0.364
FE model, re-matched ω1 = 1.000 ω1,UD
0.352
pointwise ceiling (unmanufacturable bound)
0.335
J/J_UD at held frequency